LECTURE # 9

COLOR IMAGE PROCESSING-I

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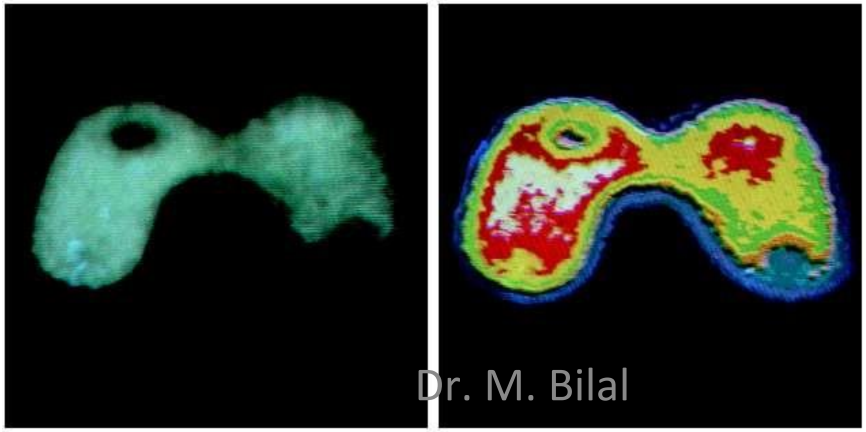
CONTENTS FOR CHAPTER 6

- Color Fundamentals
- Chromatic color
- RGB color in MATLAB
- Color Models
- RGB Model
  - **CMY**
  - **CMYK**
  - **HSI**

Color Image Processing

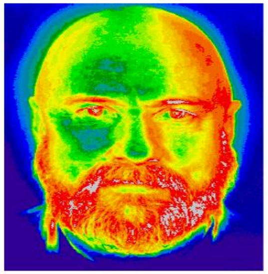
- The use of color Image Processing is motivated by two principal factors:
  - Color is a powerful descriptor
    - Object identification and extraction
    - eg. Face detection using skin colors
  - Humans can discern thousands of color shades and intensities
    - Human discern only two dozen shades of grays



Full Color Processing vs Pseudo-Color Processing

- In **Full-color Processing,** the images are acquired with a Full-Color sensor e.g. Color TV camera or Color Scanner

- In **Pseudo-color Processing,** the problem is one of assigning a color to a particular monochrome intensity or a range of intensities



COLOR FUNDAMENTALS

- **Physical** phenomenon
  - Physical nature of color is known
- **Psysio-psychological** phenomenon
  - How human brain perceive and interpret color?

Color Spectrum

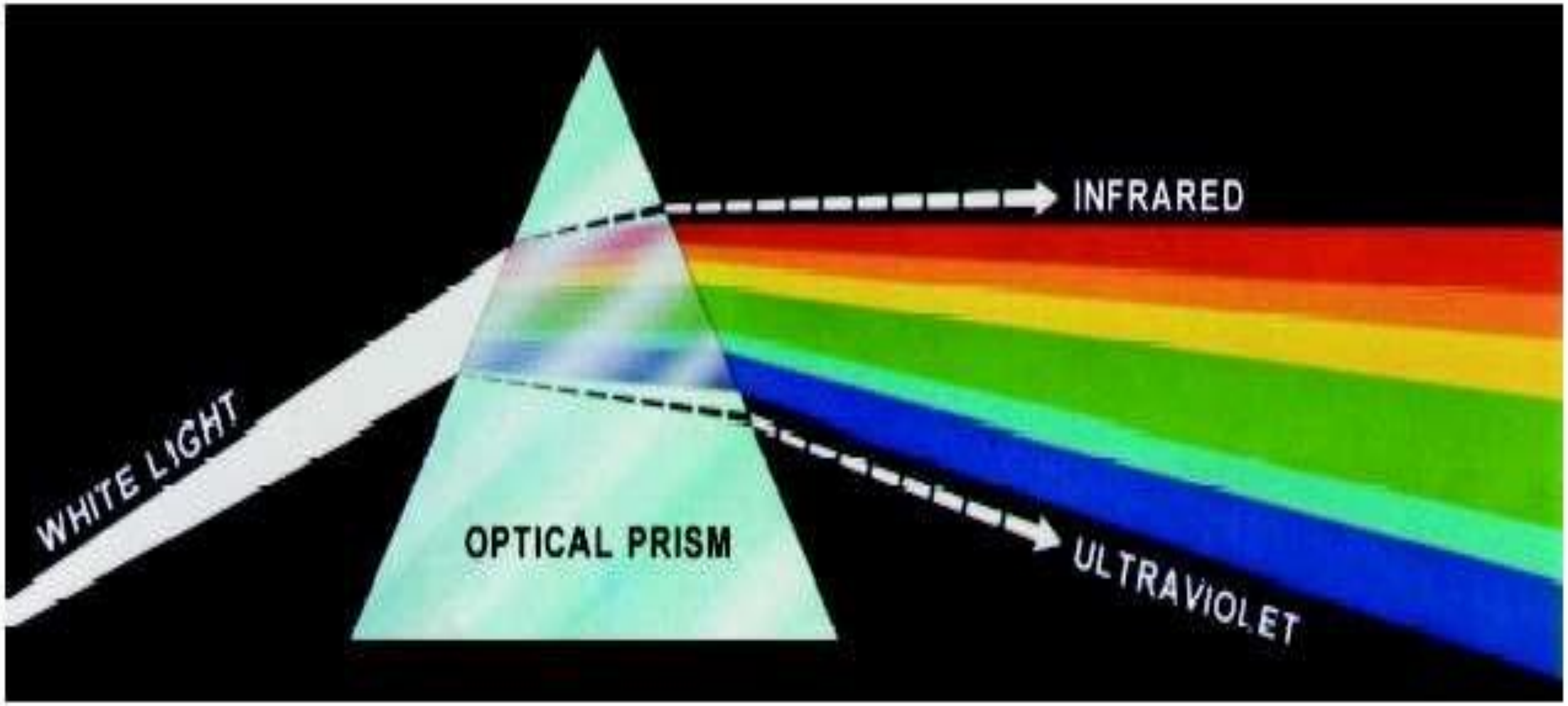


FIGURE 6.1 Color spectrum seen by passing white light through a prism. (Courtesy of the General Electric Co., Lamp Business Division.)

ELECTROMAGNETIC SPECTRUM

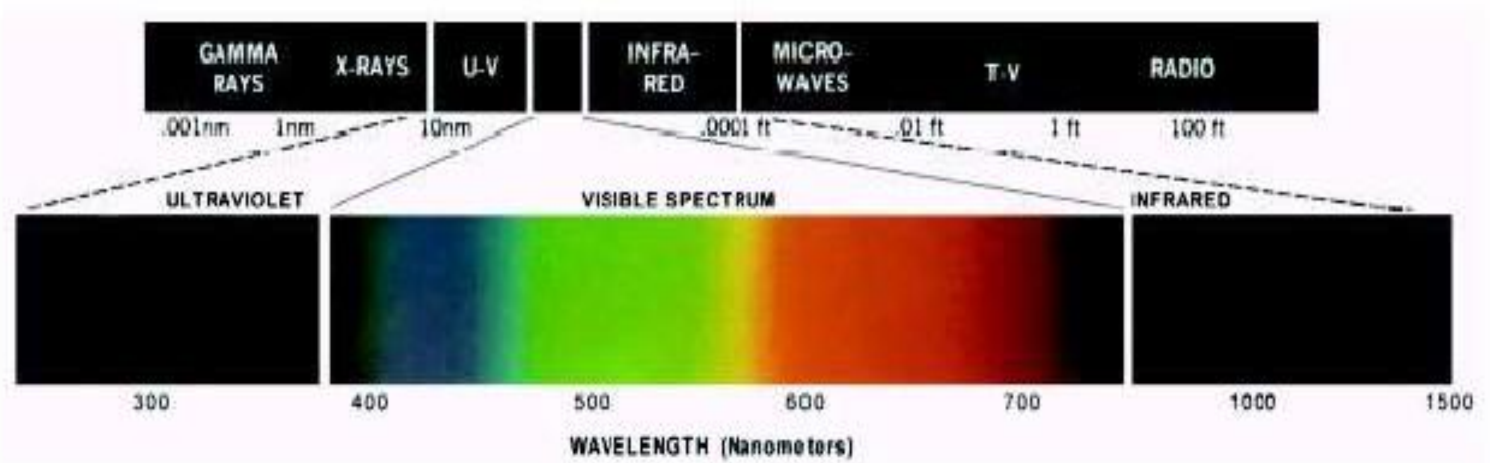
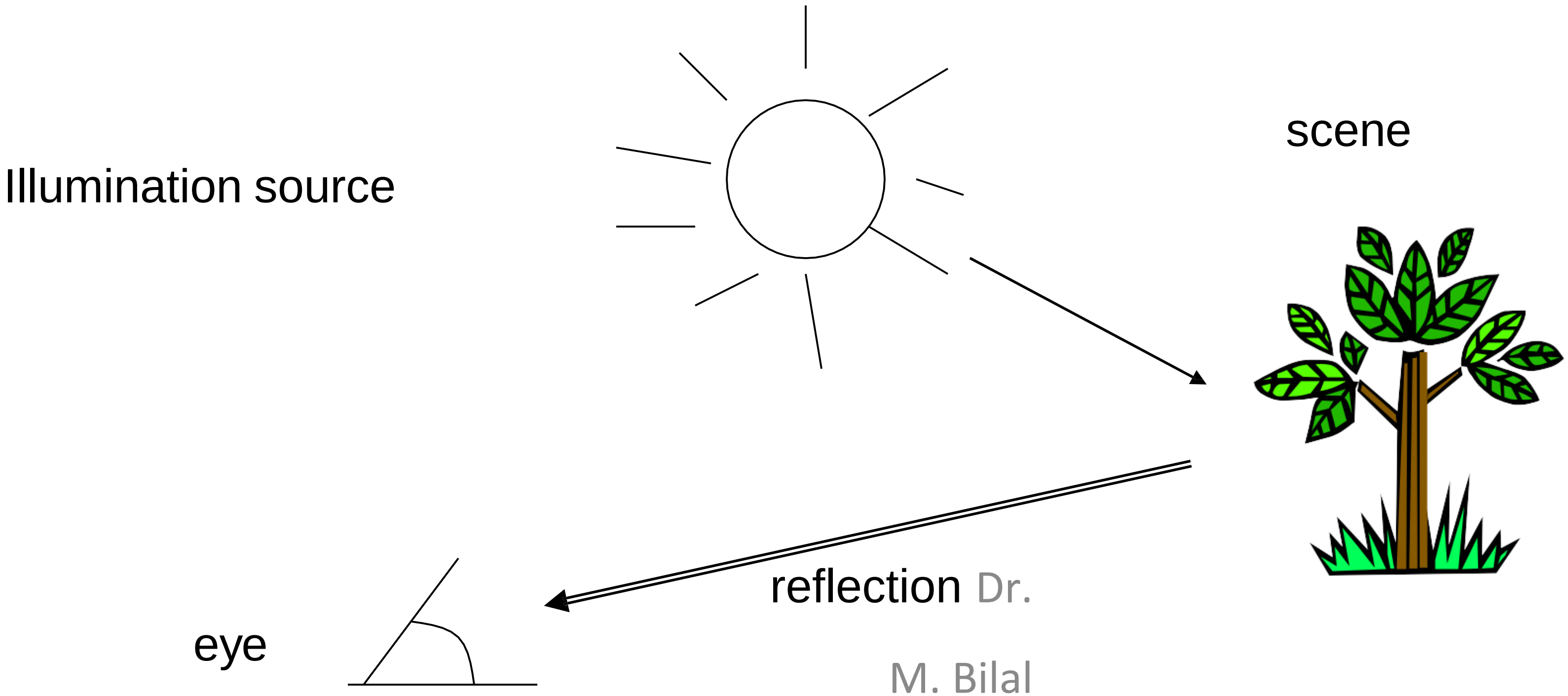
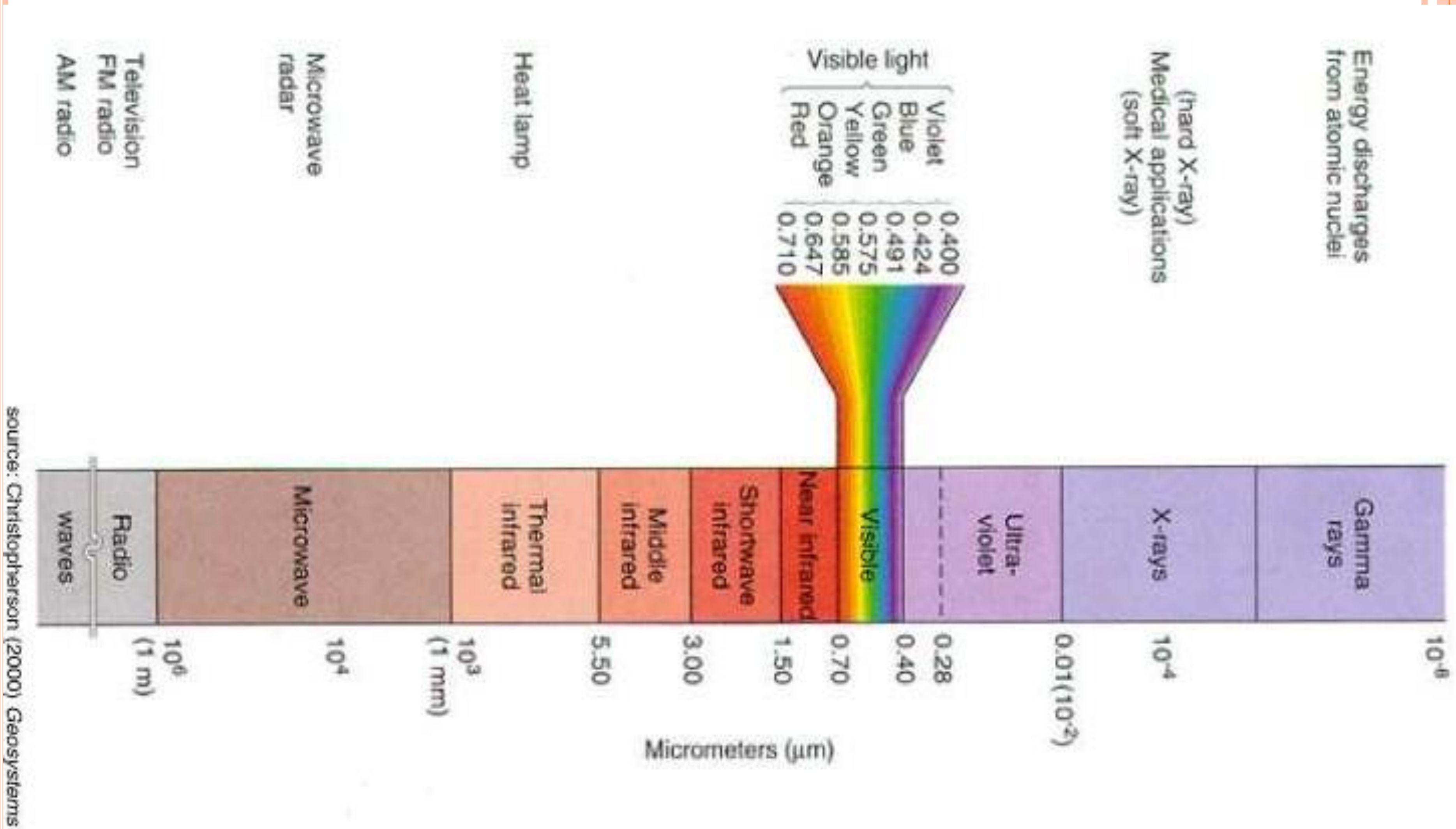


FIGURE 6.2 Wavelengths comprising the visible range of the electromagnetic spectrum. (Courtesy of the General Electric Co., Lamp Business Division.)

COLOR FUNDAMENTALS (CONT.)

- The color that human perceive in an object = the light reflected from the object




Energy discharges from atomic nuclei
Medical applications
(hard X-ray)
(soft X-ray)
Visible light
Violet 0.400
Blue 0.424
Green 0.491
Yellow 0.575
Orange 0.585
Red 0.647
0.710
Heat lamp
Microwave radar
Television
FM radio
AM radio
Gamma rays
X-rays
Ultra-violet
Visible
Near infrared
Shortwave infrared
Middle infrared
Thermal infrared
Microwave
Radio waves
10^{-6}
10^{-4}
0.01(10^{-2})
0.28
0.40
0.70
1.50
3.00
5.50
10^{3} (1 mm)
10^{4}
10^{6} (1 m)
Micrometers (μm)
source: Christopherson (2000) Geosystems

Color Fundamentals

- The colors that humans and some animals perceive in an object are determined by the nature of light reflected from the object.

Achromatic vs Chromatic Light

- Achromatic (void of color) Light: Its only contribute is its ‘Intensity’ or amount.

- Chromatic Light: spans the electromagnetic spectrum from approximately 400 to 700nm.

Quantities for Description of Quantity of Chromatic Source of Light

- Three basic quantities are used to describe the quantity of a chromatic source of light:
- **Radiance** **:** The total amount of **Energy** that flows from a Light Source

  : It is measured in **Watts**
- **Luminance :** **Luminance** gives a measure of amount of energy an observer perceives from a light source **(measured in Lumens(lm))**

  :For example, light emitted from a source operating in Infrared region of Spectrum could have significant energy (Radiance) but a human observer will hardly perceive it so luminance is zero.
- **Brightness** **:** It is a subjective measure.

  : It embodies the achromatic notion of intensity and is one of the key factors in describing color sensation

HUMAN PERCEPTION

- Detailed experimental evidences has established that the 6 to 7 million cones in the human eye can be divided into three principal sensing categories, corresponding roughly to **red**, **green** and **blue**.

- Approximately 65% of all cones are sensitive to **Red Light**, 33% are sensitive to **Green Light** and about 2% are sensitive to **Blue Light** (most sensitive).

HUMAN PERCEPTION

- Due to these absorption characteristic of Human Eye colors are seen as variable combinations of the so-called 'Primary Colors' Red, Green and Blue

- The primary colors can be added to produce secondary colors of Light

  - **Magenta** (Red+Blue)
  - **Cyan** (Green+Blue)
  - **Yellow** (Red+Green)

ABSORPTION OF LIGHT BY RED, GREEN AND BLUE CONES IN HUMAN EYE

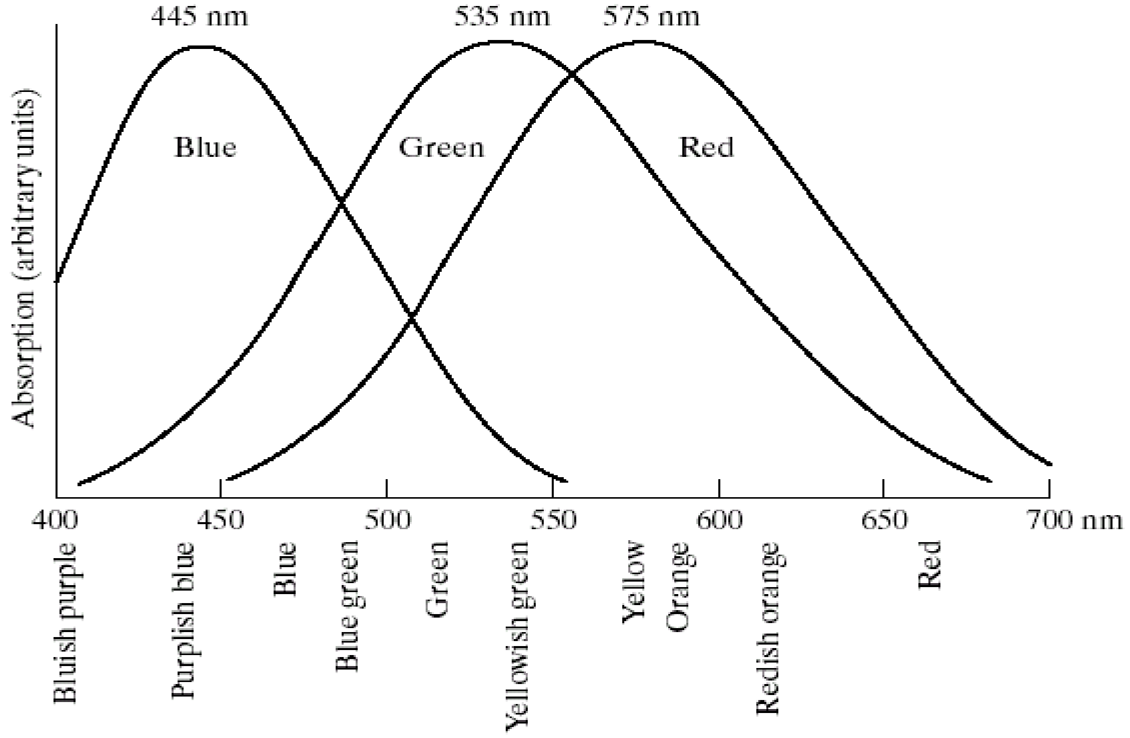
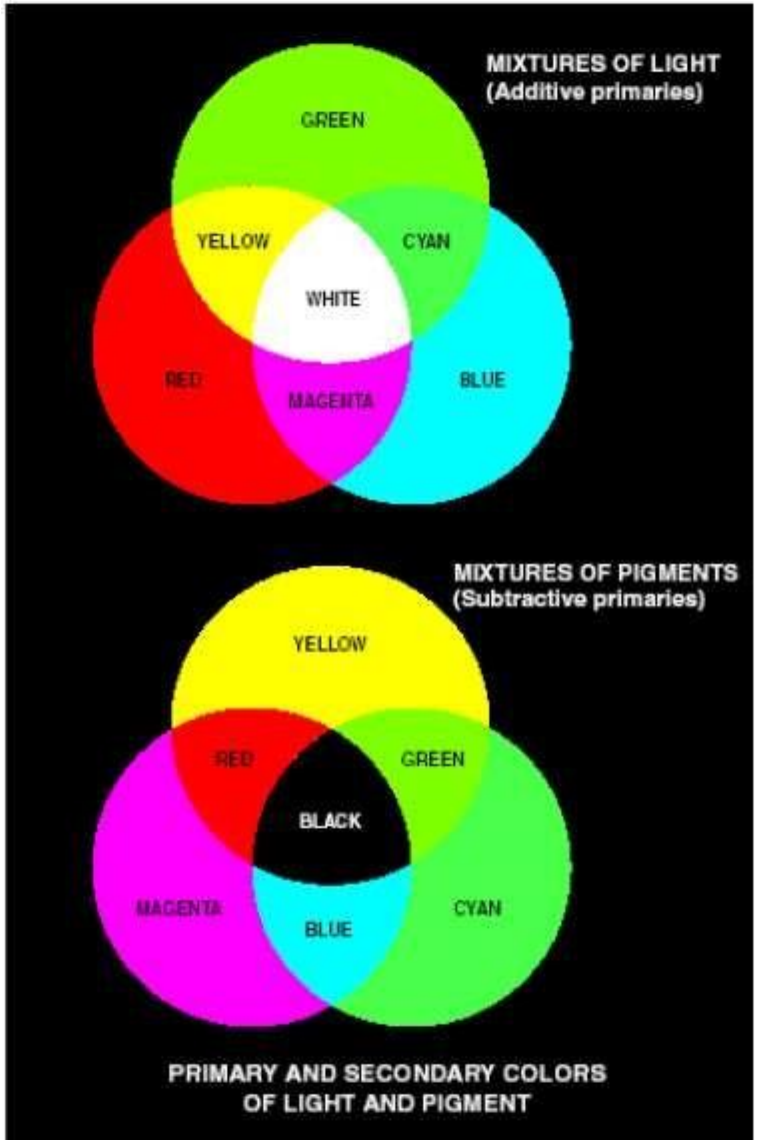


FIGURE 6.3 Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength.

- Mixing the three primaries or a secondary with its opposite primary colors in the right intensities produces white light.

PRIMARY COLOR OF LIGHT VS PRIMARY COLOR OF PIGMENTS

- Red, Green and Blue Colors are Primary Colors of Light
- In **Primary Color of Pigments** a primary color is defined as the one that subtracts or absorbs a primary color of Light and reflects or transmits the other two.
- Therefore the **Primary Colors of Pigments** are **Magenta**, **Cyan** and **Yellow** and secondary colors are Red, Green and Blue.
- A proper combination of three pigment primaries or a secondary with its opposite primary produces Black
- Color Television Reception is an example of the additive nature of Light Colors



a
b

FIGURE 6.4 Primary and secondary colors of light and pigments. (Courtesy of the General Electric Co., Lamp Business Division.)

THE CHARACTERISTICS FOR DISTINGUISHING ONE COLOR FROM ANOTHER

- Brightness : the chromatic notion of intensity
- Hue : Dominant color as perceived by an observer
- Saturation : The relative purity or the amount of white light mixed with a hue
  : The pure spectrum colors are fully saturated.
  : Colors such as pink (red and white) and lavender (violet and white) are less saturated, with the degree of saturation being inversely proportional to the amount of white light added.

Hue and Saturation taken together are called chromaticity, and, therefore, a color may be characterized by its brightness and chromaticity.

TRI-STIMULUS VALUES

- The amounts of Red, Green and Blue needed to form any particular color (denoted by X, Y and Z)

- A color is then specified by its **"Tri-chromatic Coefficients"**

$$x = \frac{X}{X+Y+Z} \qquad y = \frac{Y}{X+Y+Z} \qquad z = \frac{Z}{X+Y+Z}$$

- Thus, **x + y + z = 1**

Chromaticity Diagram

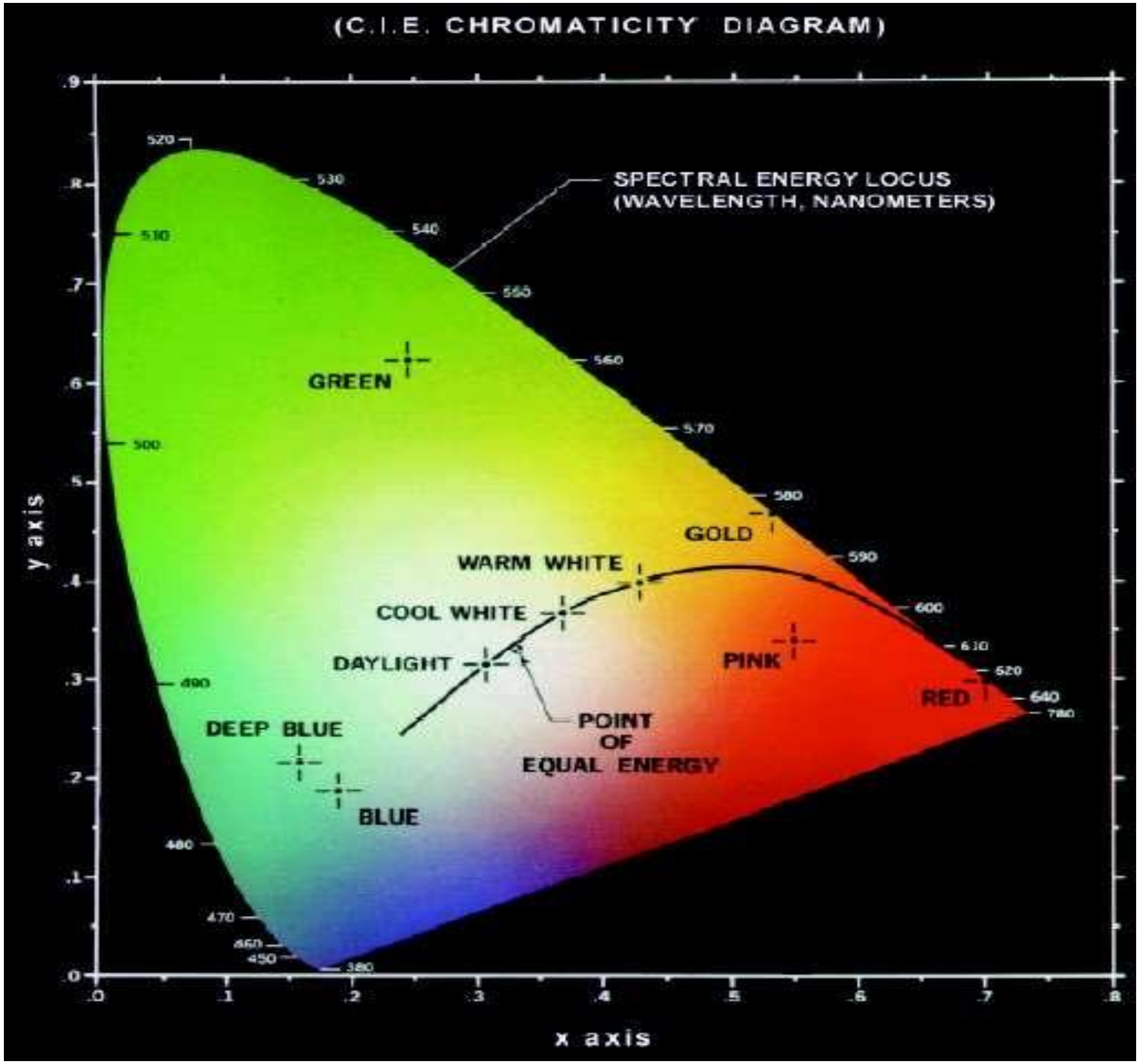
- Another approach for specifying colors is to use chromaticity diagram

- Shows color compositions as a function of x(**red**) and y(**green**)

- For any x and y the corresponding value of z(**blue**) can be obtained as

$$z = 1 - x - y$$

Chromaticity Diagram



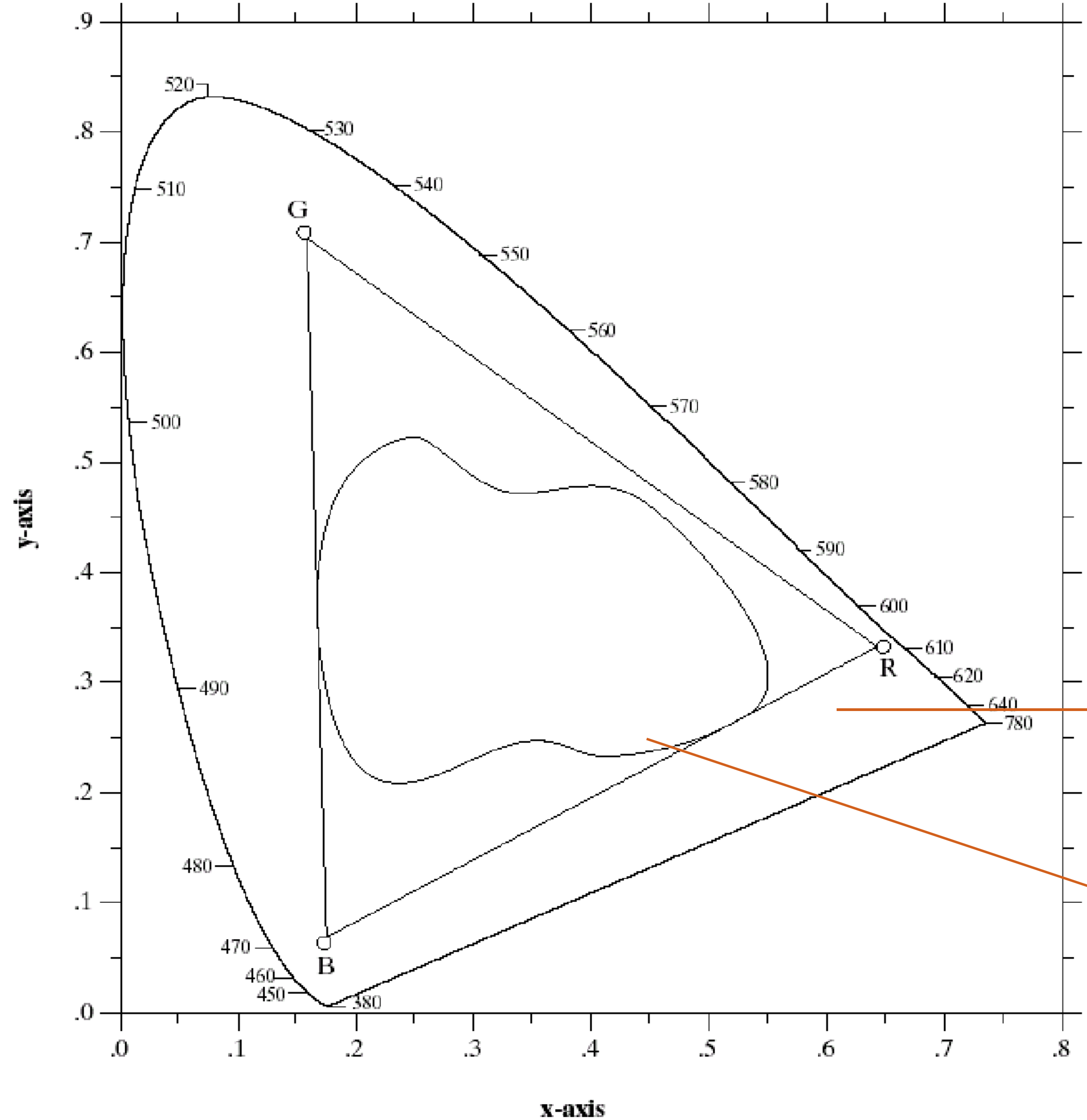


FIGURE 6.6 Typical color gamut of color monitors (triangle) and color printing devices (irregular region).

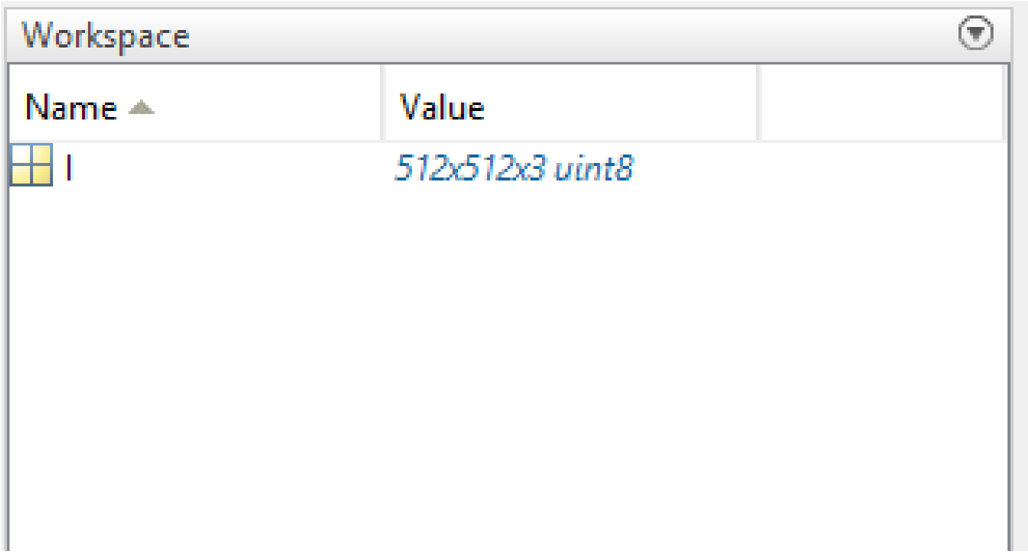
By additivity of colors: Any color inside the triangle can be produc e d by combinations of the three initial colors

RGB gamut of monitors

Color gamut of printers

RGB color image in MATLAB

- RGB color in Matlab is an 3D array
- First two index are the (x, y) location of the pixel
- Third index is the RGB pixel intensity
- E.g.



COLOR MODELS

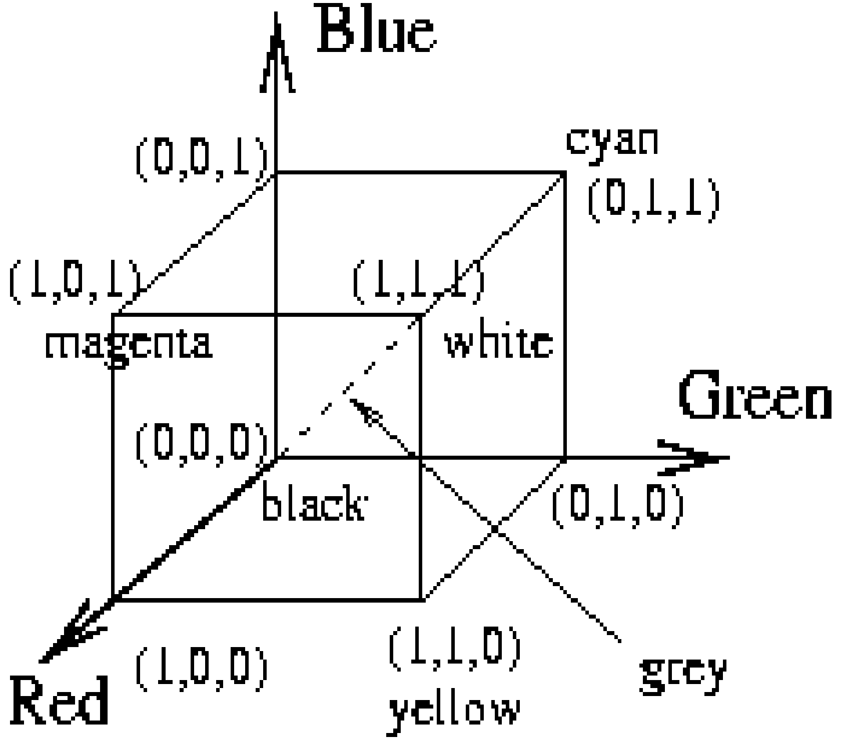
- The purpose of a color model (also called Color Space or Color System) is to facilitate the specification of colors in some standard way

- A color model is a specification of a coordinate system and a subspace within that system where each color is represented by a single point
- **Color Models**

  RGB (Red, Green, Blue)
  CMY (Cyan, Magenta, Yellow)
  CMYK (Cyan, Magenta, Yellow, Black)
  HSI (Hue, Saturation, Intensity)

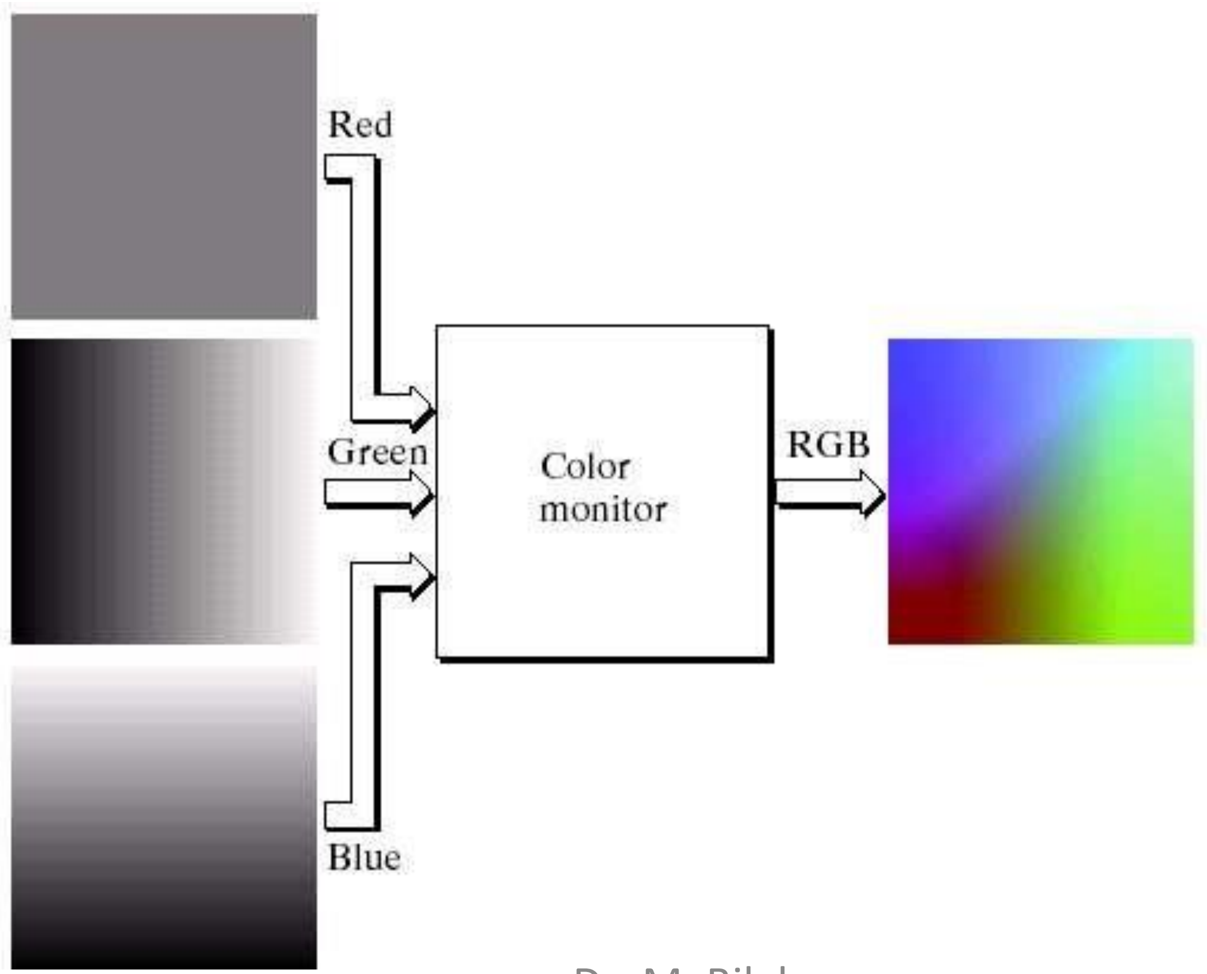
THE RGB COLOR MODEL

- Each color is represented in its primary color components Red, Green and Blue
- This model is based on Cartesian Coordinate System



APPLICATION OF ADDITIVE NATURE OF LIGHT COLORS

- Color TV



RGB COLOR CUBE

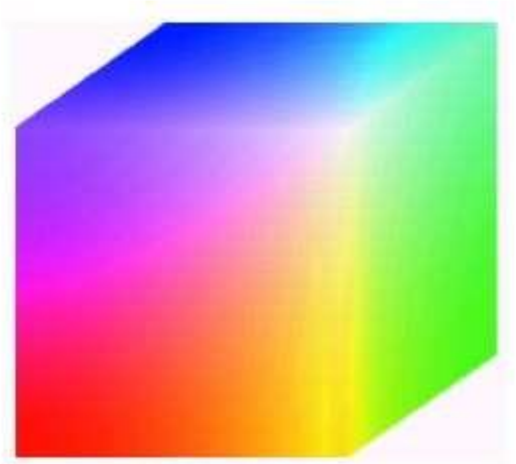


FIGURE 6.8 RGB 24-bit color cube.

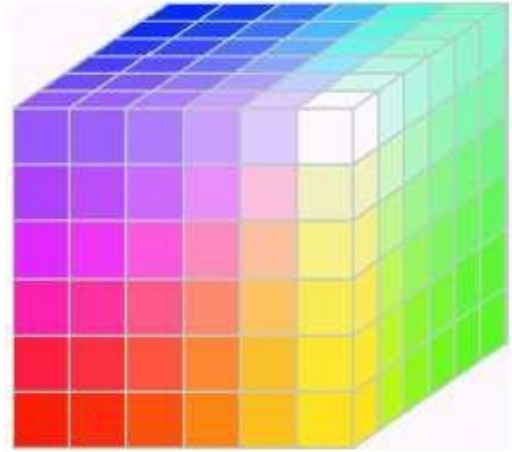


FIGURE 6.11 The RGB safe-color cube.

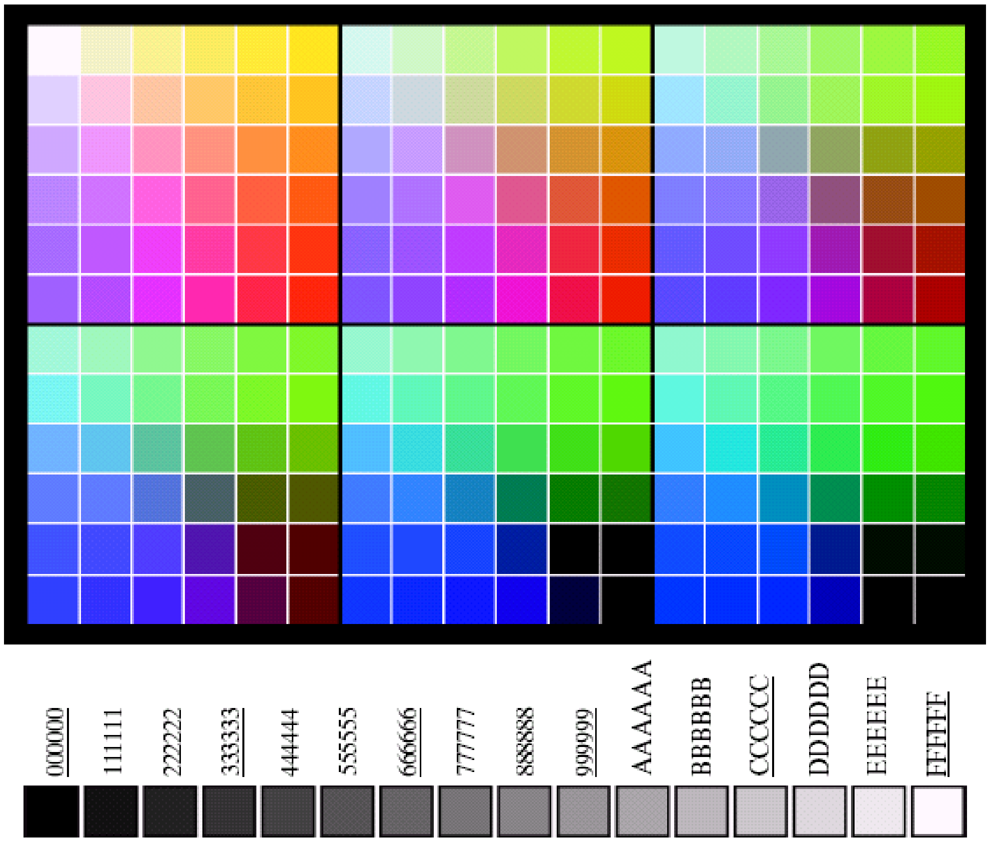
- The total number of colors in a 24 Bit image is

$$(2^8)^3 = 16{,}777{,}216 \quad (> 16 \text{ million})$$

Generating RGB image

Number System	Color Equivalents					
Hex	00	33	66	99	CC	FF
Decimal	0	51	102	153	204	255

TABLE 6.1 Valid values of each RGB component in a safe color.



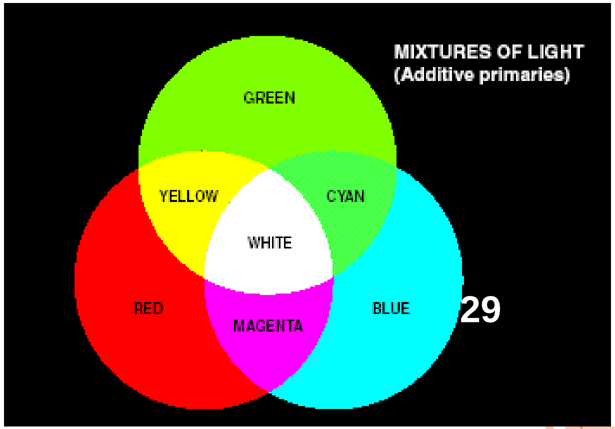
a
b

FIGURE 6.10 (a) The 216 safe RGB colors. (b) All the grays in the 256-color RGB system (grays that are part of the safe color group are shown underlined).

CMY AND CMYK COLOR MODEL

- Cyan, magenta, and yellow are the secondary colors with respect to the primary colors of red, green, and blue. However, in this subtractive model, they are the primary colors and red, green, and blue, are the secondaries. In this model, colors are formed by subtraction, where adding different pigments causes various colors not to be reflected and thus not to be seen. Here, white is the absence of colors, and black is the sum of all of them. This is generally the model used for printing.
- Most devices that deposit color pigments on paper (such as Color Printers and Copiers) requires CMY data input or perform RGB to CMY conversion internally

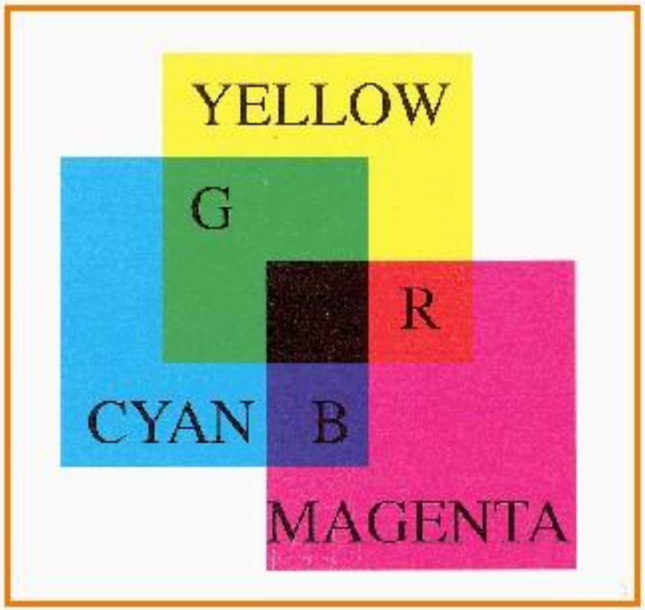
$$\begin{pmatrix} C \\ M \\ Y \end{pmatrix} = \begin{pmatrix} 1.00 \\ 1.00 \\ 1.00 \end{pmatrix} - \begin{pmatrix} R \\ G \\ B \end{pmatrix}$$



CMY and CMYK Color Model

- CMY is a Subtractive Color Model
- Equal amounts of Pigment primaries (Cyan, Magenta and Yellow) should produce Black
- In practice combining these colors for printing produces a "Muddy-Black" color
- So in order to produce "True-Black" a fourth color "Black" is added giving rise to CMYK model

CMY COLOR MODEL



Colors are subtractive

C	M	Y	Color
0.0	0.0	0.0	White
1.0	0.0	0.0	Cyan
0.0	1.0	0.0	Magenta
0.0	0.0	1.0	Yellow
1.0	1.0	0.0	Blue
1.0	0.0	1.0	Green
0.0	1.0	1.0	Red
1.0	1.0	1.0	Black
0.5	0.0	0.0	
1.0	0.5	0.5	
1.0	0.5	0.0	

RGB to CMY and CMYK

- RGB to CMY (ideal case)
  - $C = 255 - R$
  - $M = 255 - G$
  - $Y = 255 - B$

Division takes 1 or 2 instructions per bit of precision in result

- CMY to CMYK
  - $K = \min(C, M, Y)$
  - $C = 255\,(C - K) \,/\, (255 - K)$
  - $M = 255\,(M - K) \,/\, (255 - K)$
  - $Y = 255\,(Y - K) \,/\, (255 - K)$
  - 2-D lookup tables

RGB to CMYK

$K = 255 - \max(R, G, B)$

$C = 255\,(m - R) \,/\, m$

$M = 255\,(m - G) \,/\, m$

$Y = 255\,(m - B) \,/\, m$

$m = \max(R, G, B)$

- R, G, B, C, M, Y, and K have a range of [0,255]
- Useful in printers and copiers

HSI Color Model

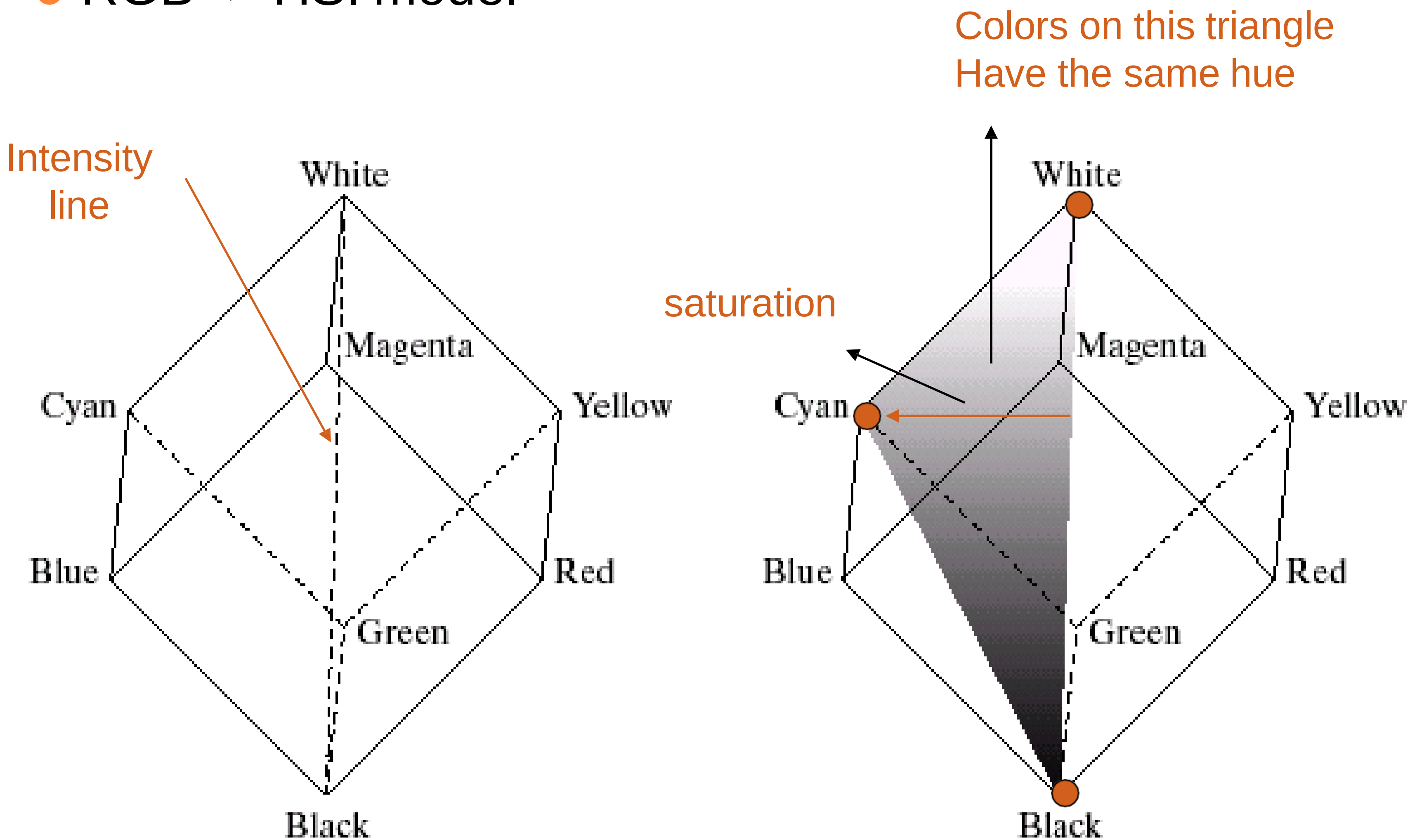
- Hue (dominant colour seen)
  - Wavelength of the pure colour observed in the signal.
  - Distinguishes red, yellow, green, etc.
  - More the 400 hues can be seen by the human eye.

- Saturation (degree of dilution)
  - Inverse of the quantity of “white” present in the signal. A pure colour has 100% saturation, the white and grey have 0% saturation.
  - Distinguishes red from pink, marine blue from royal blue, etc.
  - About 20 saturation levels are visible per hue.

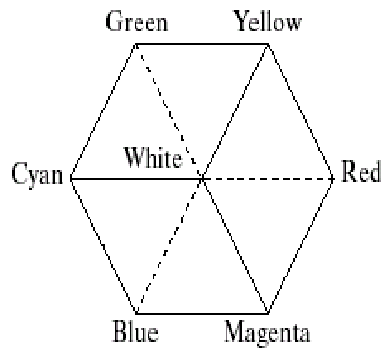
- Intensity
  - Distinguishes the gray levels.

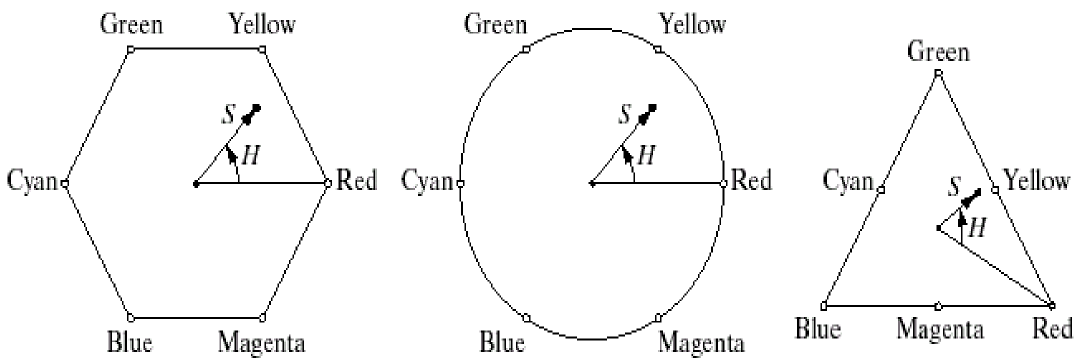
HSI COLOR MODEL (CONT'D)

- RGB -> HSI model



HSI MODEL: HUE AND SATURATION





RGB → HIS

$$I = \frac{1}{3}(R + G + B)$$

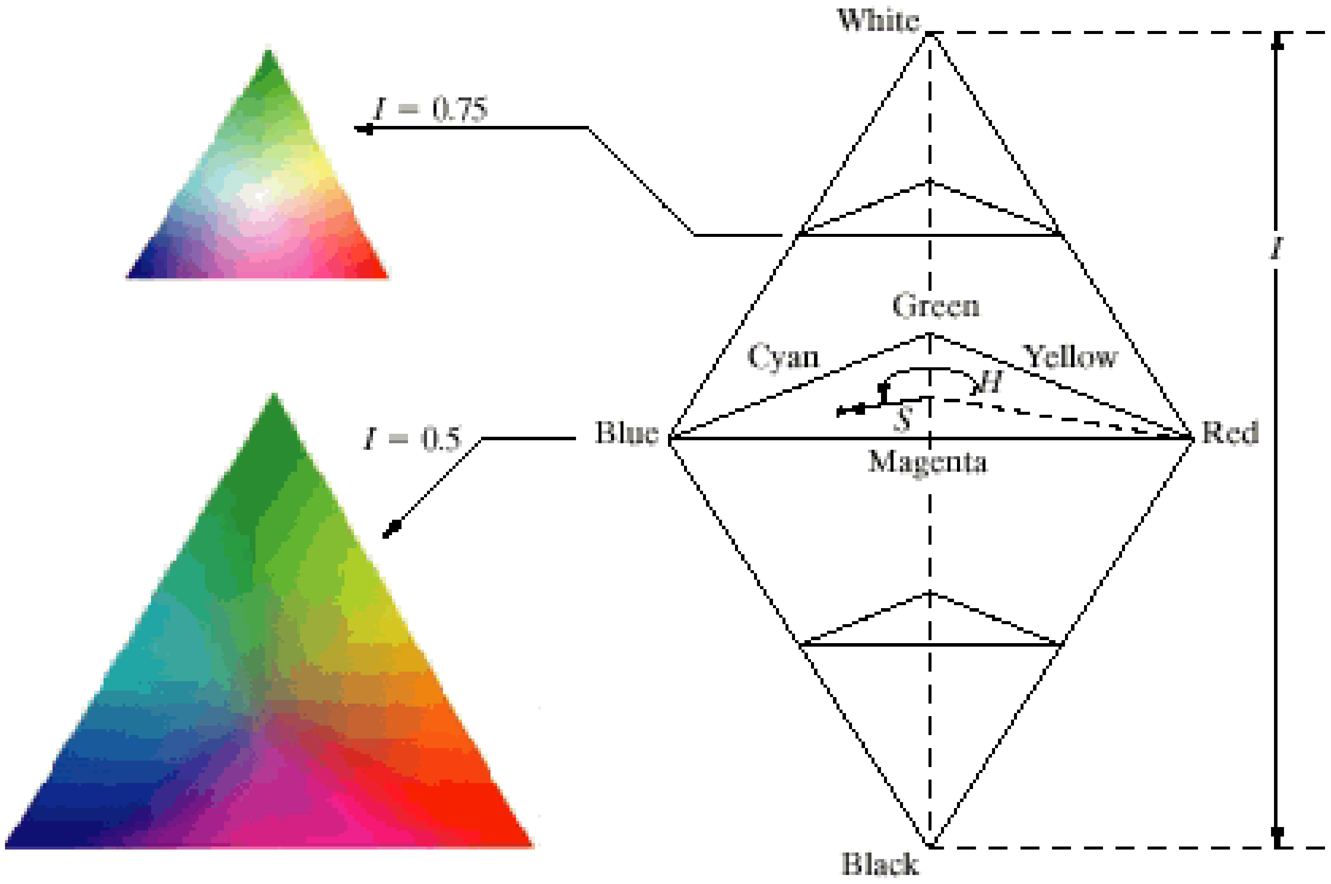
$$S = 1 - \frac{3}{(R + G + B)}[\min(R, G, B)]$$

if B≤G

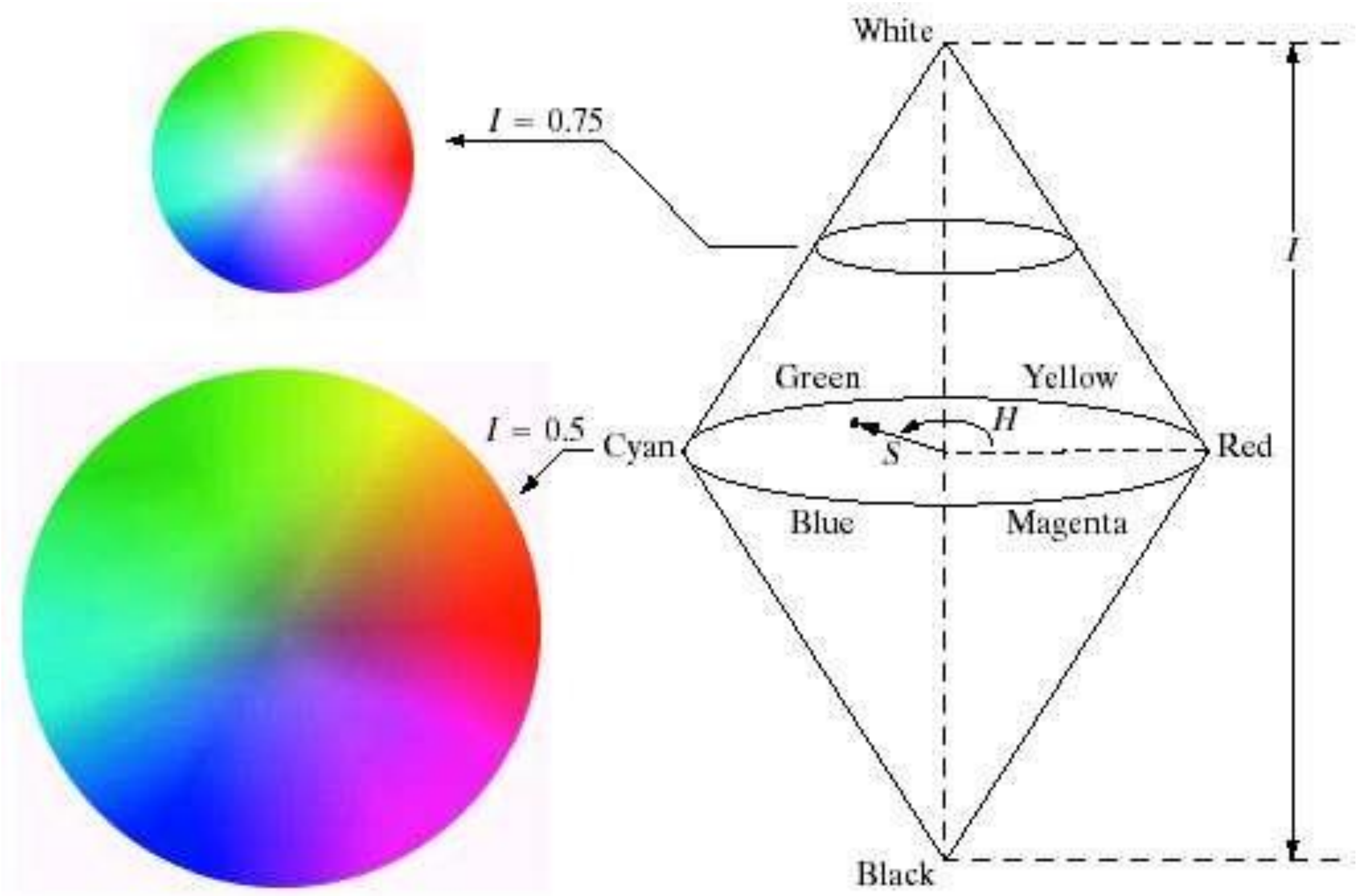
$$H = \cos^{-1}\left[\frac{\frac{1}{2}[(R - G) + (R - B)]}{\sqrt{(R - G)^2 + (R - B)(G - B)}}\right]$$

else, H=360-H

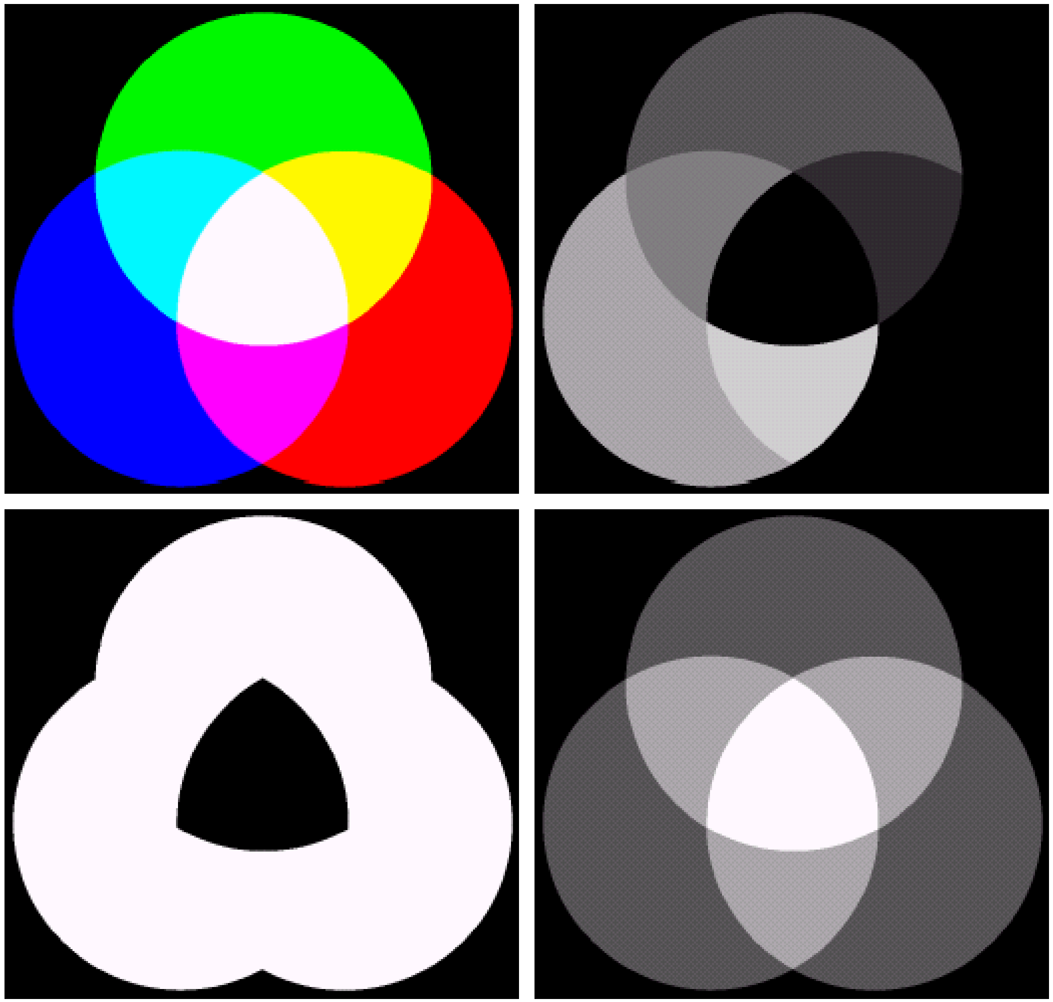
HUE, SATURATION, INTENSITY



HSI model



HSI COMPONENT IMAGES



R,G,B

Hue

saturation

intensity